Failure Analysis Report

FAIL-0096 — Synchronization Failure

1. Failure Summary

Failure ID	FAIL-0096
Failure Type	Synchronization Failure
Severity	Critical
Component	Voice Pipeline
Application	APP-009 — DeviceHub
Build Version	5.4.0
Timestamp	2024-11-16 23:43:57
Environment	ENV-007 — Staging
OS Version	Android 13
Network	LTE
Resolution	In Progress

2. Test Case Reference

Test Case ID	TC-2005
Title	Test that media streaming correctly handles invalid input when the input data co...
Execution ID	EXEC-00419
Test Run	RUN-009 — Regression Cycle 9 — Build 6.1.8
Requirement	REQ-0236 — Reliability — Device Pairing

3. Expected vs Observed Behaviour

Expected Behaviour	The system should complete playlist shuffle within the defined SLA.
Observed Behaviour	The system exhibited synchronization failure when the input data contains Unicode special characters...

4. Error Details

Error Message:

```
SyncError: conflict detected on entity 'ENTITY-2943'; last_modified timestamps diverge by 117s.
```

Stack Trace / Log Excerpt:

```
com.samsung.browser.SyncEngine.execute(line 514)
```

Log Reference	LOG-10456

5. Executed Steps at Time of Failure

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1	Re-connect the phone and confirm auto-recovery.	The fallback UI is displayed with a human-readable error message.
2	Capture a screenshot and attach as evidence.	The audit log records the event with correct metadata.
3	Verify that the UI displays 'Payment approved'.	No crash or ANR is observed.
4	Capture a screenshot and attach as evidence.	The device reconnects automatically within 10 seconds.
5	Record response time using the built-in performance monitor.	A success toast message is displayed.

6. Root Cause Analysis

OAuth2 redirect URI whitelist not updated after domain migration; authorization code flow fails.

7. Linked Defect

Defect ID	BUG-0269
Title	Synchronization Failure in Voice Pipeline: SyncError: conflict detected on entit...
Severity	Critical
Priority	Medium
Status	IN_PROGRESS
Root Cause	OAuth2 redirect URI whitelist not updated after domain migration; authorization ...
Resolution	—
Assigned To	ENG-061

8. Historical Similar Failures

HIST-0076	FAIL-0077	Synchronization Failure	6.2.0	2023-11-24	4	Open
HIST-0095	FAIL-0096	Synchronization Failure	6.5.4	2022-01-14	13	Open
HIST-0407	FAIL-0408	Synchronization Failure	4.3.0	2023-02-21	3	Resolved

9. Synthetic Data Notice

This document is entirely synthetic and intended for AI/RAG research only.

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